

Module 5: Heredity

Reproduction

Inquiry Question; How does reproduction ensure the continuity of a species

5.1.1 - Explain the mechanisms of reproduction that ensure the continuity of a species, by analysing sexual and asexual methods of reproduction in a variety of organisms, including but not limited to:

- animals: advantages of external and internal fertilisation
- plants: asexual and sexual reproduction
- fungi: budding, spores
- bacteria: binary fission
- protists: binary fission, budding

Mechanisms of reproduction

- The process by which organisms replicate themselves
- This means making a copy, a likeness, and thereby providing for the continued existence of a species
- May be asexual (no gametes transfer) or sexual (gametes transfer)

Asexual reproduction

- Only one parent needed
- No fusion of sex cells (gametes) - genetic materials don't mix
- Offspring are genetically identical clones of the parent
- Uses mitosis to create parent clones, both in unicellular and multicellular organisms
- Offspring only inherit the genes of that parent
- Almost never changes the number of chromosomes
- Forms include;
 - Fission
 - Budding
 - Fragmentation
 - Vegetative reproduction
 - Spore formation (sporogenesis)
 - Parthenogenesis

Methods of asexual reproduction;

Fission

- Parent organism is replaced by two daughters as it splits in two
- Occurs in many single-celled organisms (i.e, bacteria and fungi)

Budding

- Split resulting in a 'mother' and smaller 'daughter' offspring
- Some multicellular organisms that do this include jellyfish and hydra
 - Hydra; buds grow into fully matured individual that breaks away from parent organism

Vegetative reproduction

- New individuals formed without seeds or spores
- I.e, miniaturised plants (plantlets) on specialised leaves, and some produce new plants out of rhizomes (i.e, strawberries)
- Can also form bulbs or tubers (i.e, tulip bulbs), and some from cuttings (i.e, pineapples and potatoes)

Spore formation (sporogenesis)

- Spores form during their biological life
- Spores grow into a multicellular organism (called gametophytes in plants) without a fertilisation event
- Can be done by fungi and some algae

Fragmentation

- New organism grows from fragments of the parent
- Fragment develops into mature, fully-grown individual

- Seen in animals (sea stars), fungi and plants

Parthenogenesis

- An unfertilised egg develops into a new individual
- It occurs naturally in many plants, invertebrates (water fleas, some ants, bees, and parasitic wasps) and vertebrates (some reptiles and amphibians)
- Aphids; sexual and asexual reproduction - eggs / live nymphs at different times - typically parthenogenetic and viviparous - offspring resembles parents except for size and called virginopara

Sexual Reproduction

- Gametes are produced in sex organs by meiosis
- Male gamete (sperm) and female gamete (ovum) = zygote
- The zygote is not identical to either parent
- The zygote grows into an embryo and develops into a new individual by repeated mitosis

Asexual Analysis

- No gametes
- Only one parent = identical offspring

Asexual Advantages

- Quick reproduction means there is no need to find a mate
 - Necessary as plants = immobile
- Competitive advantage if well-adapted
- More common among plants in harsh environments, where organisms are so specific there is little benefit in population variation
- In these environments, organisms can quickly and effectively reproduce when favourable conditions arise
- Selective pressures that make asexual reproduction more effective include:
 - Food/resource shortages - asexual reproduction uses less energy
 - Small mating population/time/other mating constraints

Asexual Disadvantages

- Little/no variation = vulnerability to sudden environment changes, i.e, drought, disease, new parasite, or predator
 - May result in the survival of few, if any

Sexual Analysis

- Involves the meeting of gametes which carry genetic information from parents to offspring
- Thus, offspring contain a mix of genes, introducing genetic variation into the population

Sexual Advantages

- Genetic diversity
- Random variations may make offspring better suited to new/changing environmental conditions
- May out-compete their parents/other individuals in the population -> gaining a selective advantage

- Better chance of survival in the event of environmental change

Sexual Disadvantages

- Demands more time and energy -> courtship, finding a mate, gamete production and mating, all occur before reproduction
- The above process may increase vulnerability to predators
- This means reproduction is often the first functional process sacrificed during hardships
- Species become functionally extinct without mates
- Populations slow to build in favourable conditions

Asexual reproduction in plants - Structures

- New individuals arise from portions of the roots, stems, leaves or buds of adult individuals and are genetically identical to the parent. This type of asexual reproductive process is called vegetative propagation
- Some adult plants produce vegetative organs (i.e, bulbs, tubers, rhizomes and suckers, from which new plants can arise. This is equivalent to cloning adult plants)

Perennating Organs

- Underground organs (roots, stems, etc.) that have enough food to sustain the plant in a dormant state between seasons
- Allows the plant to survive adverse conditions (extreme cold/drought, etc.)
- Parts of the plant above ground may die, and underground organs develop buds that grow in favourable conditions
- Buds on deciduous trees are considered organs of perennation
- Gives rise to new plants when separated, making it asexual
- Gardeners exploit this by splitting bulbs, cutting rhizomes or runners, taking cuttings of shoots from stems to grow new plants
- Vegetative propagation techniques often used in agriculture - perennial crops; seedless grapes, watermelons, mangoes

Naturally occurring examples of vegetative propagation

- Runners; long, thin modified stems that grow along soil surface (i.e., strawberries and spinifex grass). At each node, leaves and roots are produced so the runner can be subdivided into new plants
- Rhizomes; are underground, horizontal modified stems, and are characteristics of ginger, ferns (i.e., bracken fern) and many grasses. It can give rise to a new shoot at each node. Gardeners often propagate ferns by splitting rhizomes
- Suckers (or sprouts); modified roots which give rise to new plants. Trees and shrubs that sucker (i.e., reeds, wattles, and blackberries) can spread into a vacant habitat after disturbance. These grow into separate plants if the parent shrub dies. Allows for rapid regrowth (after bushfire/drought)
- Apomixis; is a form of reproduction where some plants can produce offspring with generative tissues without fertilisation or seed production. Tissue may be in gamete form (i.e., unfertilised ovules) or non-reproductive (i.e., leaf tissue). It is termed generative tissue because it causes plantlets that can produce asexual seeds. Seen in plants i.e, kangaroo grass, lemon/orange trees, and dandelions

Land Plants (Embryophytes)

- Embryophyta = most familiar vegetation forming green plant subkingdom
- Includes hornworts, liverworts, mosses, ferns, lycophytes and flowering plants - not green algae (NOT A LAND PLANT)
- Complex multicellular eukaryotes with specialised reproductive organs
- Can be divided into 2 groups - seedless and seed plants
- Seedless - Bryophytes (non-vascular) and pteridophytes (vascular)
- Produce enclosed reproductive structures (gametangia and sporangia), reproducing via spores
- Seeded - Spermatophytes
- First appeared at the end of the Paleozoic era and reproduced with desiccation-resistant capsules called seeds
- 2 groups with living members:
 - > Gymnosperms (cycads, conifers, ginkgo, and gnetophytes), ovules/seeds are not further enclosed
 - > Angiosperms (flowering plants), ovules/seeds enclosed in ovaries
- Cones (conifers) or flowers (angiosperms) = male/female reproduction
- Female gametes are produced in ovaries, inside ovules, and each ovule = 1 female gamete
- Sexual reproduction in plants requires 2 different sex cells which may come from an individual organism or two parents of different sexes
- Gametes fuse to form a zygote which divides repeatedly to form new organisms
- Organisms with male and female reproductive organs and produce both sex cells are called hermaphrodites
- Some hermaphrodites are able to self-pollinate, however, others are able to recognise their own pollen and cannot reproduce with it

Pollination

- Plants rely on wind/animals as they can't move
- Plants enlist pollinators (animals)
- Self-pollination; when pollen from anther is deposited on the stigma of the same flower, or another flower on the same plant
- Cross-pollination; from 1 flower to another, increases variation and survival rate

Pollination Method - Wind

- Angiosperms; wind-pollinated and small, greenish, odourless, reduced/absent petals
- Flowers grouped together in large numbers and may hang in tassels that blow in the wind and freely shed pollen
- Many Australian grasses are wind-pollinated
- Anthers = long and produce lots of light pollen
- Stigmas are usually large and spread out like feathers to trap wind-carried pollen
- Wind pollination is inefficient, so lots of pollen is produced
- Different protein grain structures between species ensure compatibility within species

Pollination Method - Animals

- Animal-attracting flowers are more effective in ensuring pollen transfer and their reproductive success
- Adaptations in flowers are advantageous because a 1-1 plant-animal relationship

- reduces pollen wastage by ensuring it is deposited on the correct flower
- Animals (i.e, insects, birds and mammals) that act as pollinators search for rewards (i.e, nectar or pollen meals). During this, pollen rubs onto their bodies and is transferred to the next flower they visit.
- The scent, colour, markings, shape and nectar all influence what/if animals are attracted

Comparison of wind, bird, and insect-pollinated flowers

Wind

- Stamens release lots of pollen into the air; wind carries to the receptor plant (i.e, Casuarina)

Insects

- Attracted to flowers by colour and nectar, move from flower to flower (i.e, Wattles attracting beetles, wasps and bees)

Birds

- Red flowers often attract birds and plants/flowers, are strengthened to support (i.e, Grevilleas from the Proteacea family is pollinated by honeyeater birds attracted to the nectar)

Mammals

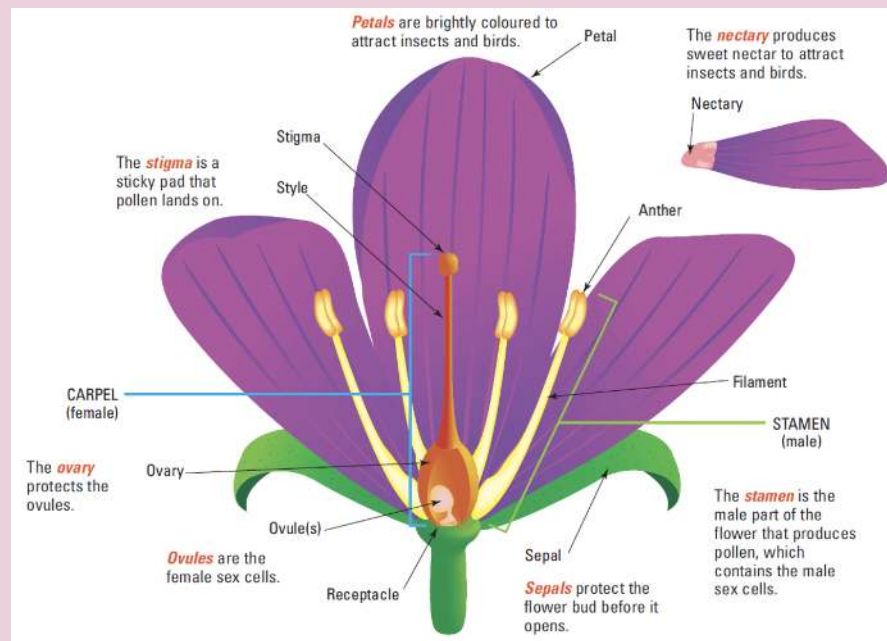
- Small marsupials are attracted, and plants/flowers are strengthened to support (i.e, Banksias attracting honey possums)

FEATURE OF FLOWER	WIND-POLLINATED FLOWERS
Petals	Small and inconspicuous, usually green or dull in colour
Scent	Usually absent
Nectar	None
Anthers	Anthers protrude outside the flower, so pollen is easily blown off by the wind; abundant pollen is produced
Stigma	Stigma protrudes from the flower, is often long, feathery and sticky, to increase surface area for trapping wind-borne pollen
Pollen	Very small grains, light and powdery; large amounts produced

BIRD-POLLINATED FLOWERS	INSECT-POLLINATED FLOWERS
Usually large and colourful, red or orange, often a tubular shape, sometimes no petals at all	Usually large and colourful (yellow or blue); may be shaped to encourage specific pollinators
Rarely fragrant because birds have little sense of smell	Often present because insects are highly attracted to scents
Large amounts of nectar produced in nectary at base of flower	Sometimes produced at base of petals, so insect must enter the flower to reach the nectar
Anthers are commonly lower than stigma, colourful, and may not be enclosed by petals	Enclosed within flower, commonly lower than stigma
Higher than anthers, sometimes not enclosed by petals and often colourful	Enclosed within flower, sticky, and commonly higher than the anthers
Sticky or powdery pollen; small amount produced	Relatively large grains and often sticky; small amount produced

Fertilisation

- After pollination -> 2 gametes form to create a zygote
- Gametes must be released at the same time, properly formed, fully developed, and with the perfect environment
- Once pollen lands on the stigma of the flower, pollen tubes grow down the style



Eggs, embryos, seeds and fruit

- Once the egg cell is fertilised by the pollen nucleus, another event sequence follows
- The fertilised egg (in the middle of the ovule) divides into a ball of cells that becomes an embryo
- Special tissue called endosperm surrounds the embryo and supplies it with food
- Ovule becomes a seed and tissue forms around it to provide a protective seed coat
- During the formation of the seed, the ovary expands and turns into a fruit

Sexual Reproduction in Animals

- Female and male sex cells must combine to form a zygote which is the first cell of the offspring
- Fertilisation can occur in or outside the body of animals
- Growth or development of the zygote (after fertilisation) can also occur in or outside the body

Gametes

- Sperm = male, too small to see
- Eggs/ova = female, roughly the size of sand grain in humans, but size varies with animal (i.e, for ostriches their eggs are around 15cm)

External Fertilisation

- Usually in aquatic sexually-reproducing organisms
- Eggs and sperm released into the water and fertilised outside the body are risky because;
 - Gametes require specific conditions and may not ever connect
 - Zygotes that do form may be eaten, battered by strong tides, or killed by pollutants

	in the water Internal Fertilisation • Sperm cell fuses with ovum inside the body (almost always female) • The result of sex in most species • Higher survival rates than external - The environment is more stable • After internal fertilisation, the zygote can be expelled as an egg or stay inside the female body • Fish, reptiles, amphibians, birds and monotreme mammals all lay eggs -> the offspring develops inside them • Amphibian and fish eggs have no shell (appear jelly-like) because they aren't at risk of dehydration, but land eggs require a shell for moisture • Eggs contain nutrient supply while oxygen can pass through the shell • Animal emerges from the shell when it outgrows the shell or runs out of nutrients • Some animals (i.e., turtles) lay eggs and leave them unsupervised until hatching. Other animals (i.e, birds) sit on eggs to keep warm and protect them • Mammals have evolved protection strategies; - Marsupials; jelly-bean-sized at birth - Not fully developed but can crawl into the pouch and attach to a teat and continue to grow and develop - Joey takes trips out of the pouch for increasingly long periods until it grows too big and the mother does not let it back in • Placental mammals (i.e, humans) keep babies in the uterus inside their bodies • A structure called the placenta is involved in providing nutrients and oxygen to the developing baby and getting rid of the baby's waste • All mammals produce milk to nourish their babies after birth
5.1.2 - analyse the features of fertilisation, implantation and hormonal control of pregnancy and birth in mammals	Mammals • Vertebrate animal of the class Mammalia • Characterised by the presence of milk-producing mammary glands for feeding their young, a neocortex region of the brain, fur or hair, and three middle ear bones. - Characteristics distinguish them from reptiles and birds, from which their ancestors diverged in the Carboniferous Period over 300 million years ago • The basic mammalian body type is a quadruped, and most mammals use their four extremities for terrestrial locomotion; but in some, the extremities are adapted for life at sea, in the air, in trees, underground, or on two legs • Main groups of mammals: - Monotremes - Placentals - Marsupials • Class <i>Mammalia</i> • Subclass <i>Prototheria</i>: monotremes: echidnas and the platypus • Subclass <i>Theriiiformes</i>: live-bearing mammals and their prehistoric relatives • Infraclass <i>Holotheria</i>: modern live-bearing mammals and their prehistoric relatives • Supercohort <i>Theria</i>: live-bearing mammals • Cohort <i>Marsupialia</i>: marsupials • Cohort <i>Placentalia</i>: placentals

Comparing Mammal Groups

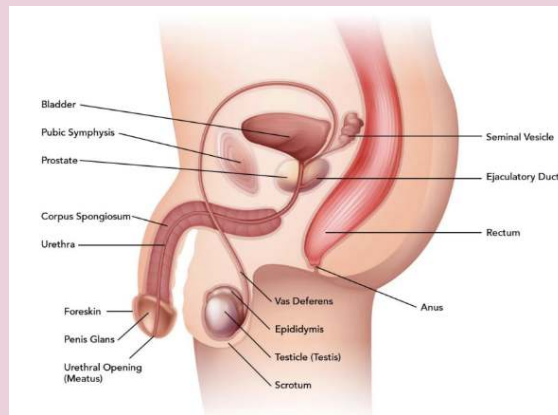
- Placentals
 - The uterus provides nourishment and protection, via a placenta and umbilical cord, for the developing embryo and foetus until birth
 - After birth, placental babies are nourished with milk and develop a covering of fur
- Marsupials
 - Underdeveloped joey is protected and nourished in the external pouch after early birth
 - This allows another fertilisation to occur internally
 - Both placentals and marsupials are **viviparous** meaning they give birth to live young
- Monotremes
 - Female lays eggs and each puggle (baby monotreme) develops inside a leathery eggshell, then hatches to be protected and fed milk by the mother
 - Only 2 monotremes; platypus and echidna
 - Monotremes are **oviparous** meaning they lay eggs in which their young develop

Strategies to maximise reproductive success

- Reproductive strategies include:
 - Internal fertilisation to increase the likelihood that gametes will meet (occurs in all 3 subclasses of mammals)
 - Implantation of the embryo into the uterine wall, with the internal development of the embryo (in marsupials and mammals) to increase the embryo's chance of survival
 - Pregnancy allows the developing young to be protected from the external environment, have a constant nutrient supply and complete a gestation period (short in marsupials; prolonged in placentals, whose young are well-developed when born)

Human male reproductive system

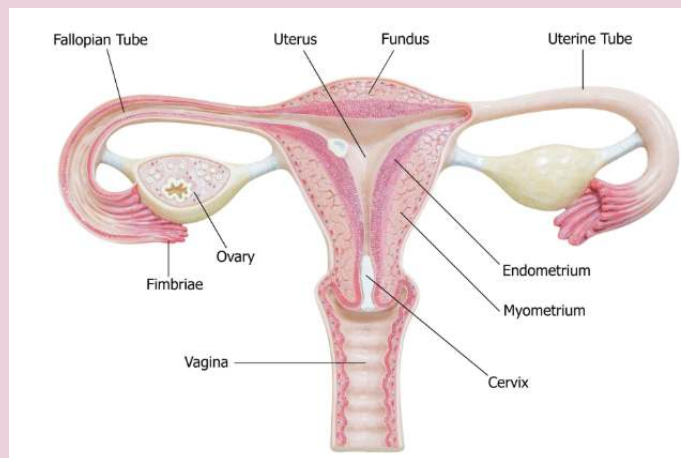
- Paired testes inside the scrotum, which produce and store mature sperm
- Accessory glands which produce secretion that makes up 95% of semen, including the **prostate, seminal vesicles, and Cowper's gland**
- A paired system of ducts called **Vas deferens** leads from the testes to the urethra
- Penis - the urethra passes through and has both sexual and excretory functions. The penis becomes erect when ready for copulation by increasing blood flow into spongy tissue until rigid
- Hormones; **lutensising hormone LH** (pituitary gland in brain) which stimulates the secretion of male steroid hormone **testosterone** (testes)



Human female reproductive system

Consists of:

- Uterus where a fertilised egg implants in the uterine wall, a placenta forms and the foetus develops until birth. The uterus undergoes changes controlled by hormones
- Paired **ovaries** that hold **oocytes** (immature egg cells) until puberty when monthly ovulation starts under hormonal control
- Paired **fallopian tubes (oviducts)** connect each ovary to the uterus. **Fimbria** at the open ends of the tube catch the eggs when they are released. Fertilisation often occurs high in the oviduct
- The **cervix** connects the uterus and vagina (which can dilate to at least 10 cm during birth). During menstruation it is controlled by the **hormone oestrogen**.
- Vagina is a muscle lined canal from the cervix to outer genitals, which receives the penis during sexual intercourse. The vagina is also the birth canal

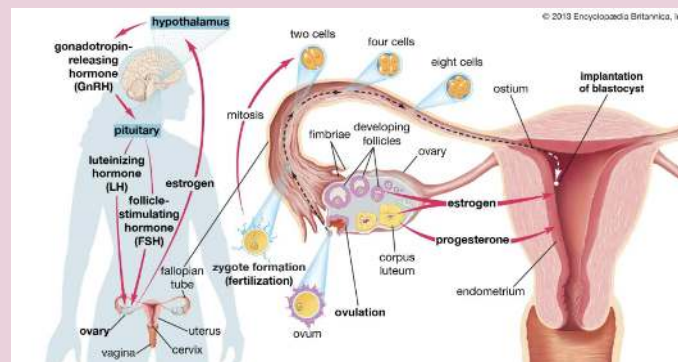
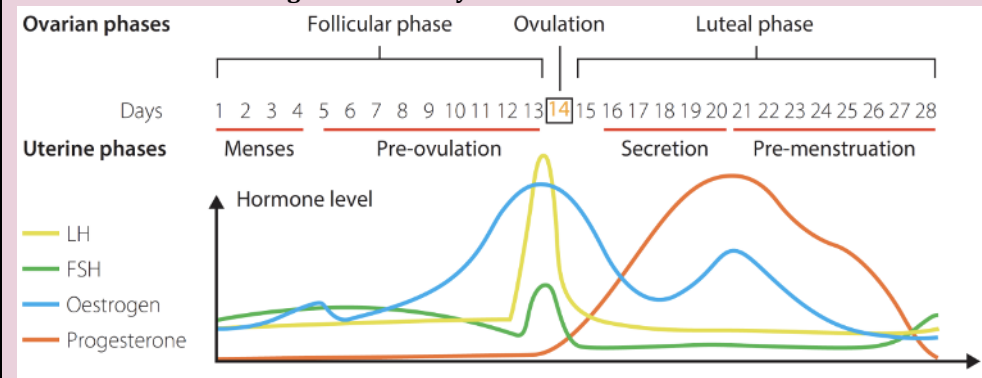


Ovulation

- Female born with all immature eggs in ovaries - first stage of meiosis (Meiosis I) begins before birth but is then paused

- After reaching maturity (puberty), ovarian cycles commence Meiosis I complete under the control of **Follicle Stimulating Hormone FSH**, and Meiosis II commences before being paused.
- Menopause; when females cease to ovulate
- Follicles containing a mature egg and nutritive cells release the hormone oestrogen which affects the lining of the uterus and also acts on the anterior pituitary gland. The egg is released when the anterior pituitary gland releases **luteinising hormone LH**
- The now burst follicle (corpus luteum) in the ovary is stimulated by LH and in turn releases both **oestrogen** and **progesterone**, which further thicken the lining of the uterus.
- If the egg is not fertilised the corpus luteum disintegrates and stops releasing hormones, the uterus lining breaks down and menstruation occurs

Female hormones through menstrual cycle



Fertilisation

- The fusion of two haploid gametes (egg and sperm) to form a single diploid zygote cell
- The zygote contains the genetic material of both the egg and the sperm. There are equal genetic contributions from each parent
- In mammals fertilisation occurs internally following mating and most often takes place in the upper part of the oviduct

- The male inserts his penis into the female's vagina and a muscular contraction (ejaculation) pushes semen from his urethra into her vagina
- From the vagina the sperm swim using their flagella, through the cervix into the uterus and then into an oviduct until it reaches and penetrates the egg

Fertilisation steps

Similar for all types of mammals

1. The sperm uses enzymes from the acrosome to dissolve and penetrate the protective layer (zona pellucida) surrounding the egg to reach the cell membrane
2. Molecules on the sperm surface bind to receptors (specialised proteins) on the egg's cell membrane to ensure that a sperm of the same species fertilises the egg, then the nucleus of the sperm enters the cytoplasm in the egg cell.
3. Changes at the surface of the egg occur to prevent the entry of multiple sperm nuclei into the egg
4. Fusion of the haploid egg and sperm nuclei results in a diploid zygote cell (the fertilised egg)

Implantation

- After fertilisation, the zygote continues to travel down the oviduct until it reaches the uterus
- The process of embryonic development begins with a stage called cleavage and when it becomes a blastocyst and reaches the uterus it is ready for implantation

Stage of development from zygote to implantation

Cleavage (days 1-4)

- Period of rapid cell proliferation by mitosis taking the zygote from a single cell to a ball of 8 cells

Morula (days 4-5)

- A ball of 16 or more undifferentiated embryonic stem cells

Blastocyst (days 5-8)

- Mitotic divisions continue, the morula becomes a blastocyst as the cells begin to differentiate. It appears like a ball with a layer of surface cells and an inner cell mass that will become the embryo

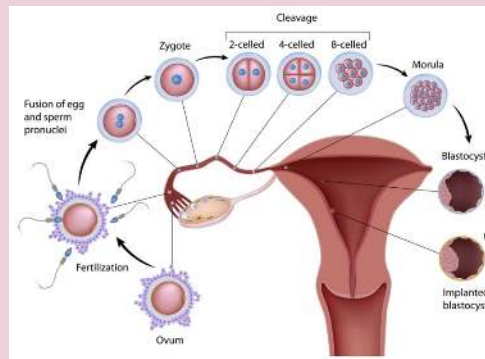
Implantation (days 8-9)

- Outer layer sends out finger-like projections into a part of the wall of the uterus (endometrium) and this area develops into the placenta

Gastrula (days 9-14)

- After the blastocyst is implanted, gastrulation occurs over approximately 5 days. In this process the blastocyst becomes a gastrula, which has 3 different layers of cells. Eventually, the gastrula becomes an embryo, then a foetus when it has formed all the basic adult features.

Ovulation to Implantation



Placenta and Umbilical Cord

- Outer layer of blastocyst initiates the formation of the placenta
- By the 5th week, an umbilical cord develops from the remnants of the egg's yolk sac and then replaces the yolk sac as the source of nutrients for the embryo, acting as a conduit for embryonic blood vessels to reach the placenta
- The umbilical cord remains attached to the foetus until after birth
- Placenta is an exchange organ bringing blood vessels from the foetus into close contact with the maternal blood
- No direct exchange of blood, rather nutrients and oxygen diffuse across into the blood of the umbilical vein and move into the foetus. The reverse happens for removal of waste products and circulation of depleted blood through the umbilical arteries from the foetus back to the placenta
- The placenta is also important source of hormones to maintain the pregnancy

Development of Embryo

- During embryonic period of development, the major organs of the body are formed from the three primary layers (germ layers) of the gastrula
 - Ectoderm
 - Mesoderm
 - Endoderm
- The germ layers are supported by two membranes; the yolk sac and the amniotic cavity
- This takes roughly eight weeks in humans (roughly 10 weeks after last period)
- At the end of the embryonic stage, the developing organism has distinct features and is known as a foetus for the remainder of its development

Germ layers

- Ectoderm (outermost layer)
 - Forms epidermis, hair, peripheral nervous system, brain and spinal cord cells
- Mesoderm (middle layer)
 - Forms muscle, cartilage, kidney and gonad cells
- Endoderm (innermost layer)
 - Forms lungs, bladder and lining of the digestive system, including the stomach, colon, liver and pancreas

Development of Foetus

- Full term pregnancy usually 40 weeks (estimated as 9 months) since the mother's last menstruation. Foetus takes 38 weeks from fertilisation to birth
- Foetus grows in size and organs continue to develop for the rest of the pregnancy (prenatal development)
- Cells become specialised to carry out their particular functions

Gestation period

- Pregnancy = gestation
- Smaller mammals usually have shorter gestation periods
- Human pregnancy divided into three trimesters (~3 months each)
 - 1st; 1-13 weeks since last menstruation
 - 2nd; 13-27 weeks
 - 3rd; 28-42 weeks (37-42 considered full term)

Average gestation period of mammals

 Mouse 20 days	 Polar bear 8 months	 Camel 13 months
 Kangaroo 1.5 months	 Human 9 months	 Giraffe 14 months
 Dog 2 months	 Cow 9.5 months	 Rhino 15 months
 Lion 3.5 months	 Seal 11 months	 Whale 18 months
 Deer 6.5 months	 Alpaca 11.5 months	 Elephant 21 months

Hormonal control of Pregnancy and Birth

- Hormones are signalling molecules that are responsible for the communication between organs and tissues to regulate physiological and behavioural processes
- Hormones are synthesised in specialised cells (endocrine glands / other tissues) and are transported to where they are needed via the circulatory or lymphatic systems or by diffusion through the extracellular fluid
- Hormones transmit signals to their target cells by altering specific biochemical reactions in these cells
- Hormones transmit signals to their target cells by altering specific biochemical reactions
- Hormones exert their effects by either directly passing through the cell membrane or indirectly by interacting with a receptor on the surface of the cell

Hormones:

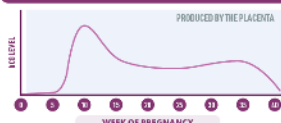
- **Progesterone** is the pregnancy hormone that prepares the uterus so that implantation and subsequent development of the foetus is sustained
- **Oestrogens** support changes during pregnancy, such as the preparation of breasts for lactation and the uterus for contractions in childbirth
- **Birth is initiated** because of stress to the foetus, the release of the **corticotrophin-releasing hormone** and the dominance of oestrogens over

progesterone

- **Oxytocin** is the main hormone of birth and causes contractions of the uterus
- **Relaxin** is a hormone that plays a role in pregnancy and birth by relaxing the pelvis and joints to allow passage of the foetus
- **Prolactin** is the main hormone needed to produce breast milk. It contributes to the enlargement of the mammary glands and prepares them for milk production. Progesterone inhibits lactation during pregnancy.
- **Human Chorionic Gonadotropin (hCG)** is produced by the placenta after implantation. It supports the function of the corpus luteum, a temporary structure in the ovaries essential in early pregnancy. It is also the hormone detected by pregnancy tests.

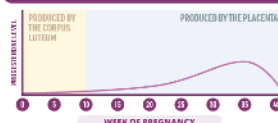
SIX KEY HORMONES IN PREGNANCY

HUMAN CHORIONIC GONADOTROPIN



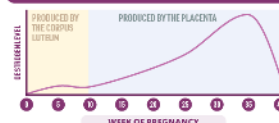
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PROGESTERONE



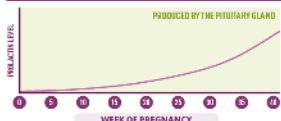
Progesterone helps establish the placenta. It stimulates growth of blood vessels that supply the womb and inhibits contraction of the uterus so it grows as the baby does. It also strengthens pelvic wall muscles for labour.

OESTROGEN



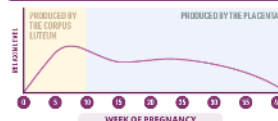
Oestrogen helps the uterus grow, maintains its lining, and helps foetal organs develop. Activates and regulates production of other hormones. With progesterone, stimulates breast growth and milk duct development.

PROLACTIN



Prolactin is the main hormone needed to produce breast milk. It contributes to enlargement of the mammary glands and prepares them for milk production. Progesterone inhibits lactation during pregnancy.

RELAXIN



Relaxin inhibits uterus contraction to prevent premature birth. It relaxes blood vessels, increasing blood flow to the placenta and kidneys. It relaxes the joints of the pelvis and softens and lengthens the cervix during birth.

OXYTOCIN



Oxytocin levels rise at the start of labour, stimulating contractions of uterine muscle. It triggers production of prostaglandins, which increase contractions further. If labour doesn't start naturally. It can be used to induce it.



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	<table><tr><th>Gland</th><th>Hormone</th><th>Hormone type</th><th>Target</th><th>Function</th></tr><tr><td rowspan="3">Anterior pituitary</td><td>Follicle stimulating hormone (FSH)</td><td>Peptide</td><td>Ovaries</td><td>Promotes development of follicle and secretion of oestrogen</td></tr><tr><td>Luteinising hormone (LH)</td><td>Peptide</td><td>Ovaries (female) Testes (male)</td><td>Promotes ovulation, development of corpus luteum, and secretion of progesterone (female), stimulates secretion of steroid hormone testosterone in testes (male)</td></tr><tr><td>Prolactin</td><td>Peptide</td><td>Mammary glands</td><td>Stimulates milk secretion.</td></tr><tr><td>Hypothalamus</td><td>Several releasing hormones</td><td>Peptides and amino acid derivatives</td><td>Anterior pituitary gland</td><td>Controls release of anterior pituitary hormones</td></tr><tr><td rowspan="2">Ovary</td><td>Oestrogen</td><td>Steroid</td><td>Reproductive tract, body generally</td><td>Promotes menstrual cycle, development of female features and behaviour</td></tr><tr><td>Progesterone</td><td>Steroid</td><td>Uterus</td><td>Prepares uterus for, and maintains pregnancy</td></tr><tr><td>Posterior pituitary</td><td>Oxytocin</td><td>Peptide</td><td>Mammary glands</td><td>Causes release of milk, stretches cervix for birth</td></tr><tr><td>Testis</td><td>Testosterone</td><td>Steroid</td><td>Reproductive tract, body generally</td><td>Development of masculine features and behaviour</td></tr><tr><td>Placenta</td><td>Human chorionic gonadotropin (hCG)</td><td>Protein</td><td>Ovaries</td><td>Maintains corpus luteum for production of progesterone, stops ovulation</td></tr></table>	Gland	Hormone	Hormone type	Target	Function	Anterior pituitary	Follicle stimulating hormone (FSH)	Peptide	Ovaries	Promotes development of follicle and secretion of oestrogen	Luteinising hormone (LH)	Peptide	Ovaries (female) Testes (male)	Promotes ovulation, development of corpus luteum, and secretion of progesterone (female), stimulates secretion of steroid hormone testosterone in testes (male)	Prolactin	Peptide	Mammary glands	Stimulates milk secretion.	Hypothalamus	Several releasing hormones	Peptides and amino acid derivatives	Anterior pituitary gland	Controls release of anterior pituitary hormones	Ovary	Oestrogen	Steroid	Reproductive tract, body generally	Promotes menstrual cycle, development of female features and behaviour	Progesterone	Steroid	Uterus	Prepares uterus for, and maintains pregnancy	Posterior pituitary	Oxytocin	Peptide	Mammary glands	Causes release of milk, stretches cervix for birth	Testis	Testosterone	Steroid	Reproductive tract, body generally	Development of masculine features and behaviour	Placenta	Human chorionic gonadotropin (hCG)	Protein	Ovaries	Maintains corpus luteum for production of progesterone, stops ovulation
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5.1.3 - evaluate the impact of scientific knowledge on the manipulation of plant and animal reproduction in agriculture	• Humans have been manipulating plant and animal reproduction for many hundreds of years • The techniques of manipulation have become more sophisticated as scientific knowledge of reproductive processes has advanced • The purpose of this human intervention is to improve the quality and yield of food • The term reproductive technology applies to any use of technology to assist and improve reproduction Early Selective Breeding • Humans have used selective breeding to produce animals and plants with more useful or more attractive characteristics • This was done without using a high level of scientific or technological knowledge. They chose those animals or plants that expressed the characteristics they wanted to conserve and selectively bred them together hoping that their offspring would show even more of these characteristics • In the past, selective breeding could only use characteristics that already existed in the genetic pool of a species Improved knowledge and technology • We now have the knowledge and skills to use cloning, recombinant DNA techniques, gene editing and transgenic technologies to transfer genes from one species to another and produce organisms with DNA combinations never seen before • We now have improved methods of control over the desired characteristics that selective breeding gives																																															

Reproductive technologies

- Reproductive technologies range from those that manipulate fertilisation (such as artificial insemination, IVF, artificial pollination) to those that split up embryonic stem cells and involve embryo implantation (including embryo splitting and cloning, as well as embryo sexing and sexing and selection)
- Reproductive technology depends on scientific knowledge and an understanding of the reproduction process, such as control of seasonal breeding, hormonal regulation of ovulation and the oestrous cycle, pregnancy and embryonic development

Plants

- Vegetative propagation techniques result in rapid expansion of genetically identical plant populations
- Selective breeding has improved the agricultural potential of many plants
- Seed banks have been developed to preserve genetic diversity
- Polyploidy (heritable condition of possessing more than two full sets of chromosomes) can be used to create seedless plants (sterile) but also to create fertile hybrids
- Cytoplasmic male sterility has been used to help ensure crossbreeding is not contaminated
- Mutation breeding (use of mutagens such as gamma rays) has been used to expand the variation in plants in the hope of producing favourable mutants for plant breeding
- Genetic modification (insertion of genes from one species into another) have been used to improve the economic value of some plants

Animals

- Selective breeding has resulted in many new strains of useful livestock
- I.E, techniques in animals include artificial insemination (AI), ovulation induction, in-vitro fertilisation (IVF), multiple ovulation and embryo transfer (MOET), juvenile in-vitro fertilisation and embryo transfer (JIVET), and somatic cell nuclear transfer cloning (SCNT)
- Genetic modification has been used in aquaculture to improve the productivity of salmon as a food source
- Sex reversal of sequentially hermaphroditic fish has been used in aquaculture to improve productivity
- Sterile triploid fish have been bred to ensure that genetically modified fish do not interbreed with wild fish

Artificial Selection + types of artificial selection

- Selective breeding by humans for desirable characteristics in plants and animals to enhance characteristics of crops and livestock
- Artificial pollination (plants)
- Transfer of pollen from an anther of a male to the stigma of a female plant by

hand

- Artificial semination (animals)
- Transfer of sperm from a male to the vagina or uterus of a female without copulation

Artificial pollination

- Practiced by Mendel in his Pea plant experiments
- Pollen is transferred from one plant using a paintbrush to the stigma of an emasculated plant
- Can be used to produce new species
- Traditional methods are often slow and unpredictable
- **How it changes genetic composition of a population:**
- Manual manipulation of pollen ensures desired characteristics will be present in offspring
- Changes in genetic composition will be widespread
- Some alleles that may be of future benefit will be lost to the population

Artificial insemination

- Lazzaro Spallanzani carried out Artificial insemination in a dog that gave birth to three offspring
- John Hunter collected sperm from an ill male and injected it into the vagina of his wife with a syringe resulting in the birth of a child in 1785
- By the 19th century it was used for livestock
- 1949 - frozen sperm increased success of Artificial Insemination
- **How it changes genetic composition of a population:**
- There is controlled breeding selecting particular characters that may not necessarily be of benefit to the animal
- Only a few stud animals may be allowed to produce offspring
- Offspring without the characteristic may not be allowed to breed
- Some alleles that may be of future benefit will be lost to the population

The Process

- Sperm is collected from the donor animal using an artificial vagina
- Sperm is checked for quality
- Sperm can either be used immediately or frozen in straws for later use
- Sperm is manually transferred to the vagina or uterus of the recipient animal

Cloning

- Cloning = process of creating a genetically identical copy of an organism
- Results in a loss of genetic diversity
- Cloning of plants has been undertaken by gardeners for years by growing plants from cuttings and grafts
- Other methods of cloning:
- Tissue culture; commonly used in plants to produce thousands of offspring, small pieces of plant tissue are grown on sterile medium and cultured for

future use

- Embryo twinning; used in animals, the embryo is split at a very early stage in development, each new identical half is allowed to proceed
- Nuclear transfer; a mature cell nucleus is combined with an egg cell (with its nucleus removed) of another animal. The resulting embryo is implanted into an animal for development

Dolly

- Product of Somatic cell nuclear transfer in 1996
- How its done:
 - Nucleus removed from unfertilised egg cell
 - Remove the nucleus from a mature somatic (body) cell
 - Transfer the nucleus from the mature cell into the egg cell
 - Stimulate the egg to divide with small electric current
 - Culture the cells until they are a ball of cells that can be implanted into a surrogate mother

Saving the Wollemi Pine

- Wollemi Pine, *Wollemia nobilis* was discovered in 1994
- Less than 100 trees in wild
- Thought to be extinct and only found as fossils dating back 90 million years
- Very little genetic variation in population - all genetically identical
- Location of the Wollemi pines in the wild is kept secret to protect them from people
- Secret plant nursery set up to propagate cuttings from trees in the wild, this then went into mass production with the aid of tissue culture
- Lots of Wollemi pines being commercially available will hopefully save the species from extinction and protect the wild population from the actions of people

Transgenic Species

- Transgenic; an organism which has been created through genetic engineering and has DNA from another species inserted into its genome
- Transgenic species initially create genetic diversity but then lead to loss of Biodiversity with large plantings of genetically identical crops producing a monoculture
- A transgenic organism is made using recombinant DNA technology

Manipulating DNA

- DNA must be manipulating to produce a transgenic species in a three-step process
 1. Restriction Enzymes cut DNA into smaller pieces at specific locations. These are also known as gene shears. If the cut ends are uneven, they are called sticky ends
 2. Ligases help make and repair DNA. When DNA has been cut with restriction

enzymes, ligases strengthen the repair bonds between the newly reformed DNA (**recombinant DNA**)

3. Once the recombinant DNA is formed, multiple copies are created using **polymerase chain reaction (PCR)**. The polymerase catalyses DNA replication to create billions of copies very rapidly

Transgenic Species Methods

Four main methods:

1. Microinjection
 - DNA from one species is inserted into a cell from another species using a fine glass needle or micropipette
2. Ti Plasmid
 - Bacterium *Agrobacterium tumefaciens* inserts its DNA into plants using a structure called a Tj Plasmid. Genetic engineers splice a desirable gene into the bacteria's plasmid and then grow the tissue in a culture
3. Biolistics
 - A gene gun fires small metal particles coated with DNA into another cell nucleus
4. Electroporation
 - Cells are exposed to short electrical impulses so that small pores form in the membrane allowing new genes to be inserted

Examples of Transgenic organisms

- OncoMouse
 - First organism to be patented
 - Genetically engineered to contain a human cancer gene, each mouse lives for eight weeks before it dies from cancer
 - Used in research programs to search for a cure for human cancer
- Bt Cotton
 - *Bacillus thuringiensis* is naturally occurring soil bacterium that produces a chemical that kills caterpillars and closely related insects
 - The gene that produces this toxic chemical can be inserted into plants such as cotton
 - The cotton can then produce its own pesticide that kills the caterpillars when they eat parts of the plant, reducing the need to spray crops and reducing costs

Transgenic Issues

- Producing transgenic species involves combinations of genes that have never existed and are unlikely to naturally occur
- Many transgenic species become food for human consumption, safety of GM foods unknown especially for people with allergies
- New pathogens could be created and released
- Super weeds could be created by the cross fertilisation of wild and transgenic species

- Transgenic animal species have basic rights

Cell Replication

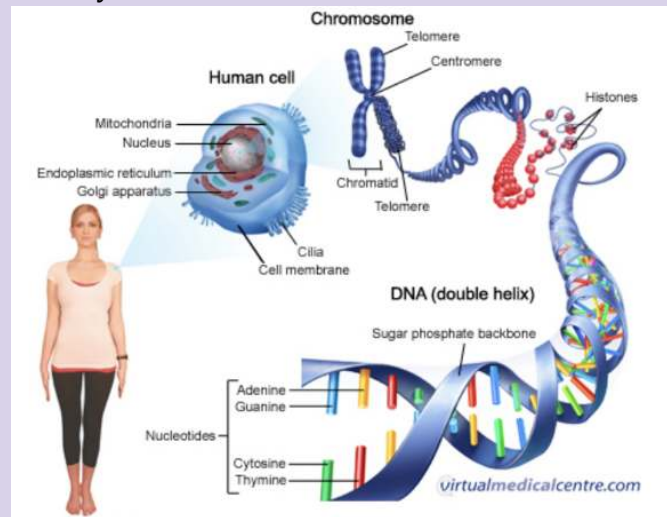
5.2.1- Model the processes involved in cell replication, including but not limited to:

- Mitosis and meiosis
- DNA replication using the Watson and Crick DNA model, including nucleotide composition, pairing and bonding

DNA and Chromosomes

- DNA - Deoxyribonucleic Acid
 - Nucleic acid containing the four bases (adenine, thymine, cytosine and guanine) which forms the major component of chromosomes and which contain coded genetic information
- Chromosome
 - Thread-like structure composed principally of DNA and visible in cells during mitosis meiosis

Where you find DNA



DNA Structure

- Made of simple repeating units called nucleotides
- Each Nucleotide has a sugar (deoxyribose) part, a phosphate part and a nitrogen containing base
- 4 nucleotides
 - 2 pyrimidines: Cytosine and Thymine
 - 2 purines: Adenine and Guanine
- Base Pairing Rule
 - Adenine + Thymine
 - Cytosine + Guanine
- Phosphate-deoxyribose backbone is composed of phosphate groups and deoxyribose sugars
- Base pairing rule can be used to construct a complimentary DNA strand
- Double stranded molecule twisted into a Double Helix

- Each DNA molecule consists of 2 nucleotide chains that run in opposite directions
- The sugar-phosphate backbones of the two chains are on the outside of the DNA double helix
- The bases (ATCG) are arranged so that they point to the inside of the molecule
- Weak Hydrogen bonds form between the base pairs

DNA Replication

- DNA is able to make exact copies of itself
- The process begins when a region of double stranded DNA unwinds to form two short lengths of single-stranded DNA
- Each single strand acts as a template for building a new complementary strand
- Results in the formation of two double stranded molecules of DNA, each containing one strand of the original DNA molecule and both identical to the original molecule

When Replication takes place

- Replication must occur before mitosis for growth and repair of body cells
- Replication is also part of the process of meiosis where gametes are formed

Mitosis

- Type of cell division that results in the replication of identical cells
- Mitosis results in growth of an organism, is involved in the healing of wounds and the replacement of cells (e.g, red blood cells and skin cells)
- All the body cells or somatic cells come from pre-existing cells by mitosis
- Mitosis occurs in areas of rapid growth in organisms
- These sites are in different places in different types of organisms. This is usually due to the need for rapid replication such as growth points or sites where repair to damaged tissue is required

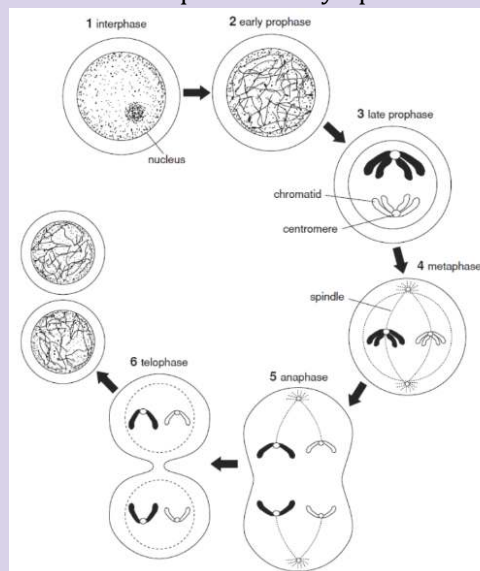
Mitosis in Action

- When a cell divides, a series of changes occur in the nucleus of cells
- Each time a cell divides by mitosis, new daughter cells end up with chromosomes, and hence DNA, which is identical to those of the original parent cell
- Mitosis is essentially the replication of chromosomes and their separation into daughter cells
- Plastids and mitochondria are self-replicating organelles. This means that when mitosis is occurring these organelles reproduce independently of the nuclear division
- Chloroplasts (a type of plastid) and mitochondria both possess genetic material (DNA) that enables them to replicate. It is thought that they may be descendants of ancient prokaryotic cells that have since become part of

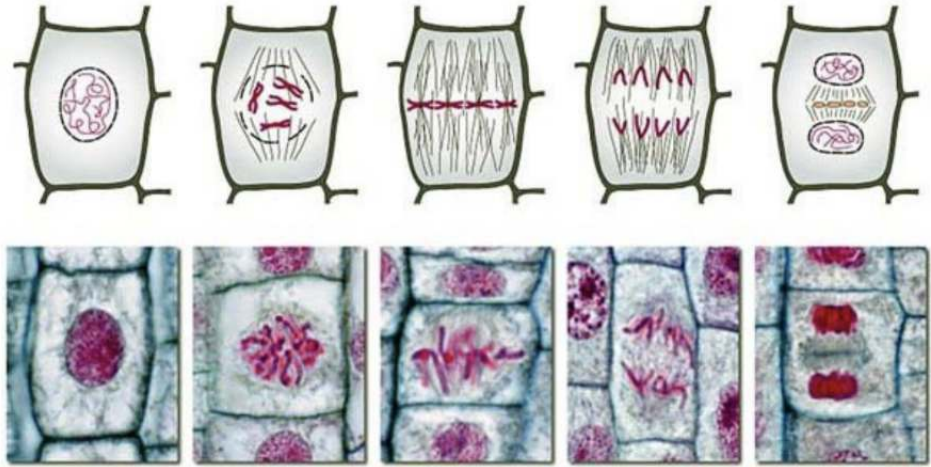
other cells.

Stages of Mitosis

- *Interphase*
 - During this stage each chromosome becomes replicated
- *Prophase*
 - Chromosomes are visible, each chromosome and its replica are held together by a structure called the centromere
- *Metaphase*
 - The chromosomes move through the cytoplasm to the spindle and become fastened to it by their centromere, the chromosomes are in the middle of the cell
- *Anaphase*
 - Daughter chromosomes move apart, each member of a pair moving to opposite poles of the cell, the cells start to become constricted in the centre
- *Telophase*
 - Nuclear membranes form around each group of daughter chromosomes. The chromosomes uncoil to become slender threads. A new cell membrane forms at the equator. The cytoplasm divides and two new daughter cells now exist



Mitosis in plant cells



Plant cells at different stages of mitosis



Cytokinesis

- After the chromosomes have separated into two nuclei the cytoplasm divides so that the cells are able to enter interphase
- Cytokinesis is important because after mitosis the nucleus has divided and separate nuclear membranes of the daughter cells have formed. The cytoplasm then must divide (cytokinesis) to produce two new cells

Meiosis

- Meiosis means reduction division, so meiosis makes haploid (n) sex cells)
- After DNA duplication occurs, the homologous pair of chromosomes can become entwined (crossing over), parts of one chromosome, including its genes, might be exchanged with its homologue
- When the pairs move apart there is an equal chance that the new cell will get either chromosome (**random segregation**)
- The chromosomes in different homologous pairs move independently into new cells (**independent assortment**)
- In the second division, haploid (n) gametes are formed

Stages of Meiosis

Meiosis I

- **Interphase:** DNA replication occurs, creating identical sister chromatids attached at the centromere
- **Prophase I:** Chromosomes condense, homologous chromosomes pair up (synapsis), and crossing over (exchange of genetic material) occurs. This stage is further divided into leptotene, zygotene, pachytene, diplotene, and diakinesis.
- **Metaphase I:** Homologous chromosome pairs align at the cell's equator.
Anaphase I: Homologous chromosomes are pulled apart to opposite poles of the cell.
- **Telophase I and Cytokinesis:** Nuclear membranes reform, chromosomes decondense, and the cell divides into two haploid daughter cells.

Meiosis II

- **Prophase II:** Chromosomes condense again, and spindle fibers form
- **Metaphase II:** Sister chromatids line up at the cell's equator
- **Anaphase II:** Sister chromatids are pulled apart to opposite poles
- **Telophase II and Cytokinesis:** Nuclear membranes reform, chromosomes decondense, and the two daughter cells divide, resulting in a total of four haploid daughter cells (gametes)

Processes influencing genetic variation in offspring

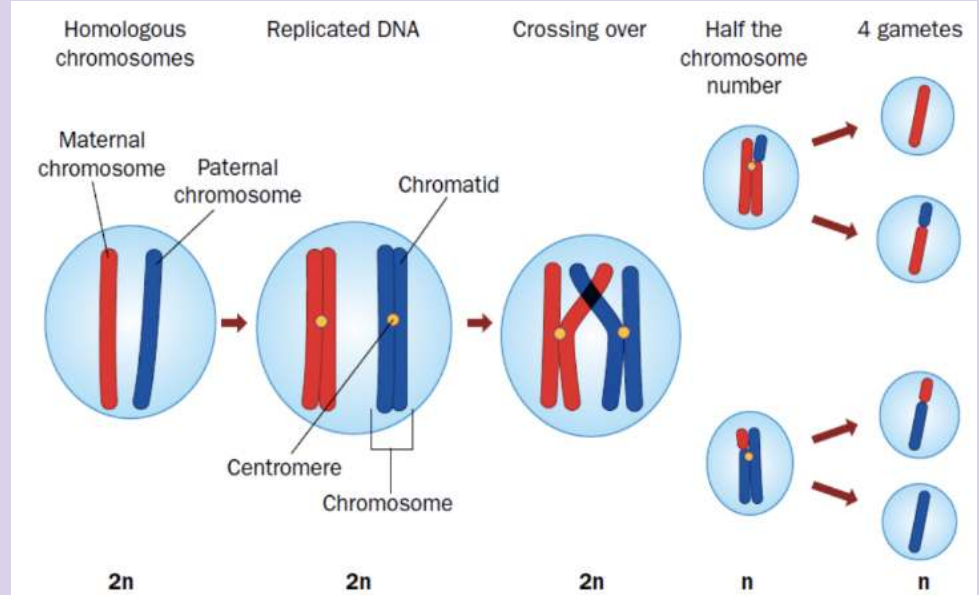
- Three processes:
 1. Crossing over
 2. Random segregation and independent assortment of homologous chromosomes
 3. Random fertilisation of the egg by a successful sperm

Chromosome behaviour during Meiosis

- During meiosis, chromosomes exhibit the following behaviour:
 - Replication of DNA

- Shortening and thickening of chromosomes
- Formation of chromosomes consisting of two chromatids joined at the centromere
- Crossing over and exchange of genetic material
- Random segregation and independent assortment of homologous chromosomes
- Formation of four gametes each with unique genetic material

Summary of chromosome behaviour



5.2.2 - Assess the effect of cell replication processes on the continuity of species

Cell Replication

- James Watson and Francis Crick did not simply come up with their model of the structure of DNA. They relied heavily upon earlier work by other scientists and researchers and by ongoing research at the time.
- Essentially, they came up with their model through educated guesswork and modelling, making many mistakes along the way. They did very little fundamental research into the structure. But with the help of Maurice Wilkins and the work of Rosalind Franklin they were able to deduce the structure of DNA

Early work on DNA structure

- 1920's Phoebus Levene
- Discovered that Deoxyribose sugar molecules of DNA are linked with phosphate groups

- Noted that nitrogen bases are attached to a deoxyribose sugar and a phosphate group
- Provided first model of a nucleotide consisting of:
 - A five carbon deoxyribose sugar
 - A phosphate group
 - One of four nitrogen bases
- 1950 Erwin Chargaff
- Measured the proportions of nitrogen bases and found they were not always in equal proportions and that the proportions were identical within a species but were different between species
- The proportion of Adenine always equalled the proportion of Thymine (same for Cytosine and Guanine); **Chargaff's Rule**
- 1953 Linus Pauling
- Stated that because proteins could be helical in structure and were held together by electrical charges or Hydrogen Bonds, perhaps DNA had a similar shape

Early DNA Models

- Alexander Todd worked out the chemical structure of DNA, with the backbone consisting of alternating sugar and phosphate groups
- Sven Furberg proposed that DNA formed a single stranded helix with the bases sticking out and stacked on top of one another
- Watson and Crick originally tried a 3-strand model of DNA with the sugar-phosphate backbones in the centre with the bases on the outside
- Linus Pauling independently came up with a 3-strand model similar to Watson and Crick
- Watson and Crick developed a 2-strand model still with the sugar-phosphate backbone in the centre
- Then Watson tries model with sugar-phosphate at outside and bases paired with identical bases to suit Chargaff's rule better
- Watson made cut-outs of bases and found that bonds between A&T, C&G fit together without fudging and they form identical shapes and allow for Chargaff's rule

Rosalind Elsie Franklin (1920-1958)

- Used X-Ray crystallography to discover that the shape of the DNA molecule was a helix (photograph 51)
- Her Photograph 51 showed that DNA was a helix shape because of the black cross reflections, which also allow measurements of the structure to be made
- X-ray crystallography is used to record the location of atoms in a crystal
- Was not recognised by the *Nobel Assembly at Karolinska Institutet* for her contributions as she had died

Maruice Hugh Frederick Wilkins (1916-2004)

- Studied the structure of large molecules
- Informed Watson and Crick of Franklin's discoveries (without her permission) which enabled them to change their model to produce a Double Helix structure
- Awarded Nobel Prize for Physiology or Medicine in 1962

James Dewey Watson (1928-)

- Worked with Crick to model the structure of the DNA molecule
- Suggested that pairing of bases made it possible to copy and pass on genetic information
- Awarded Nobel Prize for Physiology or Medicine in 1962

Francis Harry Compton Crick (1916-2004)

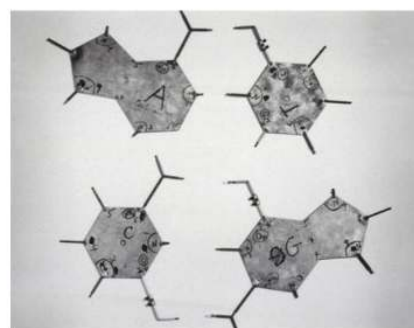
- Worked with Watson to model the structure of the DNA molecule
- Studied the genetic code
- 1962- awarded Nobel Prize for Physiology or Medicine

The Model



Original model proposed by Watson & Crick

Bases that were used in the model



DNA and Polypeptide synthesis

5.3.1 - Construct appropriate representations to model and compare the forms in which DNA exists in eukaryotes and prokaryotes

Effect of Mitosis on Continuity of Species:

- Mitosis is the driving force for the cells to grow that results on the overall growth of organisms. If it weren't for mitosis, no

	organism would have reached the ideal size and shape for it to survive in the environment. • Life of most multicellular organism begins from a single cell termed as Zygote. Without mitosis, a zygote would never be able to transform into a complete adult organism. • Maintains the equality of chromosomes in number and similarity in characteristics throughout all the cells. • In both plants and animals' mitosis cell division leads to the formation of new organs. • Mitosis helps in retaining purity of types because usually recombination and mutation does not occur during Mitosis cell division. • Helps in regenerating new tissues by replacing old ones (example: gut epithelial cells and blood cells). Effect of Meiosis on Continuity of Species: • Helps in formation of haploid gametes for carrying out sexual reproduction. • Plays a vital role in maintaining ideal number of chromosomes in organisms. • Introduces new characteristics in organisms as a result of recombination due to crossing over. • Due to meiosis cell division, when haploid gametes unite to form zygotes, the zygote possesses 50% trait from the mother and the rest 50% from the father.
5.3.2- model the process of polypeptide synthesis, including: transcription and translation, assessing the importance of mRNA and tRNA in transcription and translation, analysing the function and importance of polypeptide synthesis, and assessing how genes and environment affect	DNA- Deoxyribonucleic Acid (contains deoxyribose sugar) RNA- Ribonucleic Acid (contains ribose sugar) mRNA- messenger RNA is copied from the DNA template and is hundreds to thousands of nucleotides long and carries information to ribosomes in the cytoplasm. mRNA is single stranded, contains ribose sugar and Uracil instead of Thymine, triplets of bases are called codons tRNA- transfer RNA is copied from DNA template and is only 75-85 nucleotides long, contains anticodons which match with codons on mRNA rRNA- ribosomal RNA are the enzymes that make proteins within the ribosome Unzipping DNA

phenotypic expression

- DNA is able to unzip where the teeth of the zipper are the nitrogenous bases held together by weak electrostatic **hydrogen bonds** that are easily pulled apart under some chemical environments
- DNA unzips fully when replicating with each side acting as a template on which to build a complementary strand
- DNA partially unzips during polypeptide synthesis to make a code or message for a unit of a protein known as a polypeptide, after which it zips back together

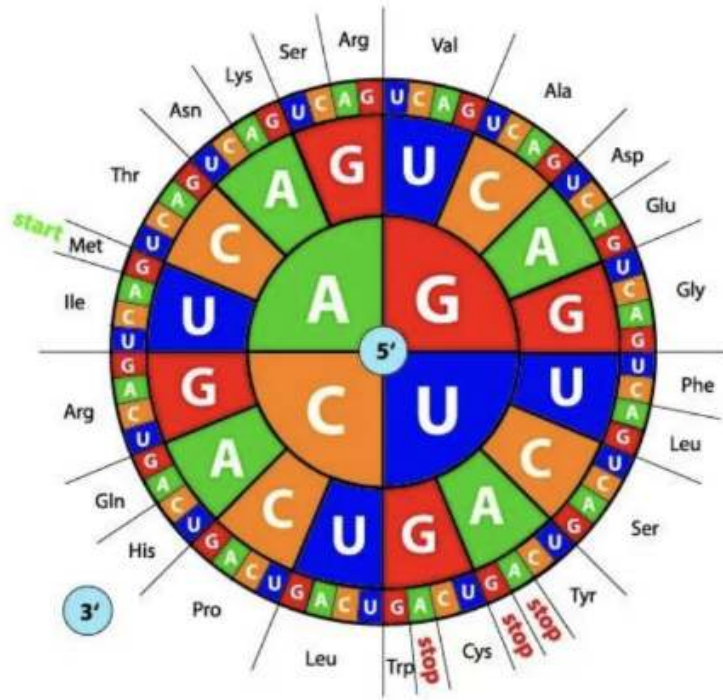
Genetic Code

- Genetic information that codes for the assembly of amino acids is stored as **3-letter codes** called **Triplets** on the DNA strand
- On the mRNA strand the equivalent of the triplet is the **codon** (also based on the three-letter code)
- Each codon represents one of 20 amino acids used in the construction of polypeptide chains
- There may be more than one codon for each amino acid

Codes for amino acids (mRNA)

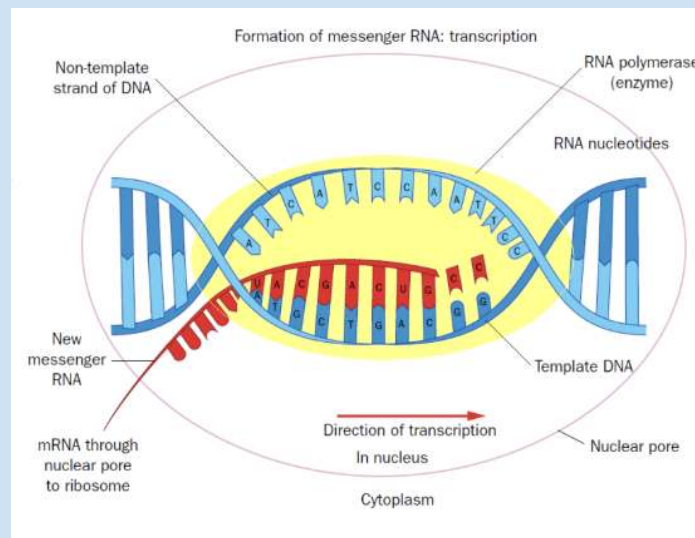
	U		C		A		G					
U	UUU } UUC }	Phenylalanine	UCU } UCC }	Serine	UAU } UAC }	Tyrosine	UGU } UGC }	Cysteine				
	UUA } UUG }		Leucine		UCA } UCG }		UAA } UAG }		Stop codon Stop codon	UGA } UGG }	Stop codon Tryptophan	
	C	CUU } CUC }		Leucine	CCU } CCC }	Proline	CAU } CAC }	Histidine		CGU } CGC }		Arginine
		CUA } CUG }	CCA } CCG }		CAA } CAG }		CGA } CGG }					
A		AUU } AUC }	Isoleucine		ACU } ACC }		Threonine	AAU } AAC }	Asparagine	AGU } AGC }	Serine	
		AUA } AUG }			Methionine or start codon			ACA } ACG }		AAA } AAG }		
	G	GUU } GUC }	Valine	GCU } GCC }		Alanine		GAU } GAC }	Aspartic acid	GGU } GGC }	Glycine	
		GUA } GUG }		GCA } GCG }	GAA } GAG }			Glutamic acid		GGA } GGG }		

Codon Wheel



Transcription - Formation of mRNA

- RNA polymerase enzyme makes single strand of messenger RNA using the template strand of DNA
- Inside the enzyme, free nucleotides pair with their corresponding base, gradually building the mRNA strand
- Uracil substitutes for Thymine in the formation of the mRNA strand
- mRNA exits nucleus through pore and enters the cytoplasm where it engages enzymes to begin translation

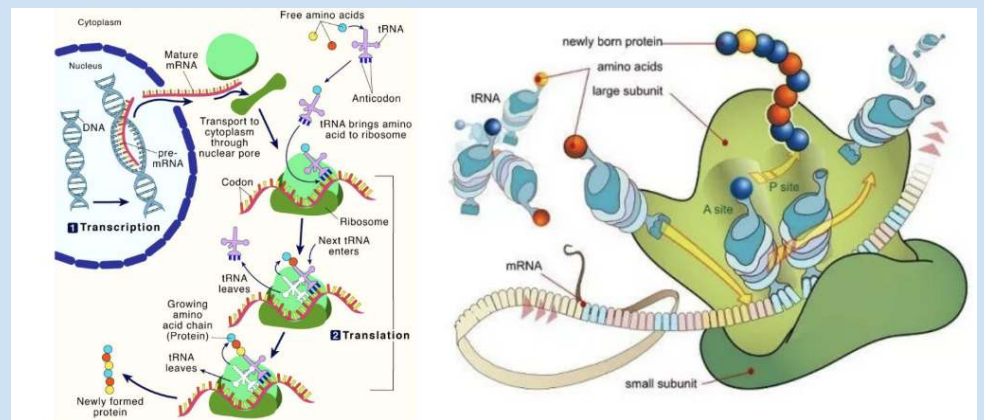


Translation - Formation of Polypeptide

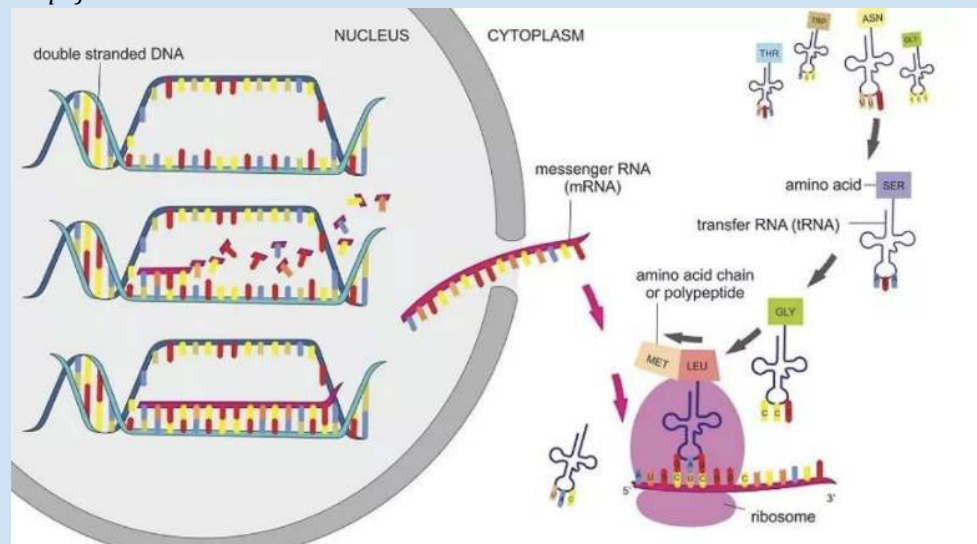
- Ribosomes, made up of a complex of ribosomal RNA (rRNA) and proteins, attach to a binding site on the mRNA molecule
- Ribosomes have binding sites that attract tRNA molecules loaded with amino acids
- There are different tRNA molecules for each of the different possible anticodons
- Once a match between the codon (on mRNA) and anticodon (on tRNA) is made, the previous tRNA molecule releases its amino acid and unbinds from the mRNA. The amino acid is attached to a growing polypeptide chain
- The unloaded tRNA molecule attracts another amino acid to its attachment site to repeat the process

Process Summary

Detailed vers:



Simplified vers:



Function and importance of polypeptide synthesis

- Provides the main component of proteins which are essential to the structure and functioning of cells, tissues and organisms
- Provides a mechanism for the action of inheritance
- Antibodies are proteins formed when polypeptides are made in response to the presence of antigens
- The figure shows that the immune response is dependent on polypeptide synthesis to protect the body from infection

Environment and Phenotype

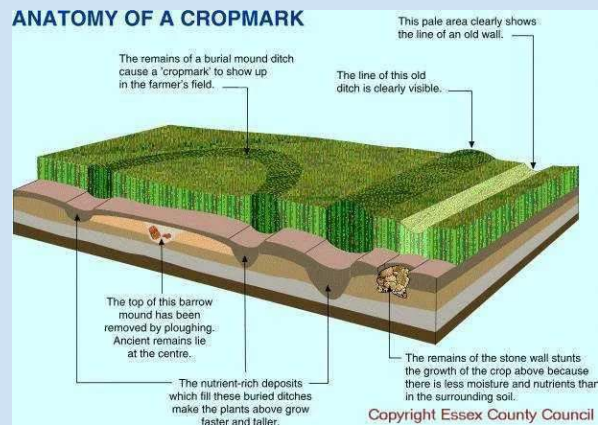
- The genetic information contained in genes is not the only influence on

phenotype

- There is a range of phenotypes possible from any genotype
- I.e, plants that are genetically identical may grow to different heights depending on the nutrients they receive, if insufficient nutrients are available, all the plants will not reach their full genetic potential

Crop Mark

- Crop marks seen from the air have been used by archaeologists to identify the remains of ancient structures and buildings that cause differential growth in otherwise uniform growing crops
- The differential growth is caused by nutrient-rich areas promoting growth and nutrient deficient areas retarding growth



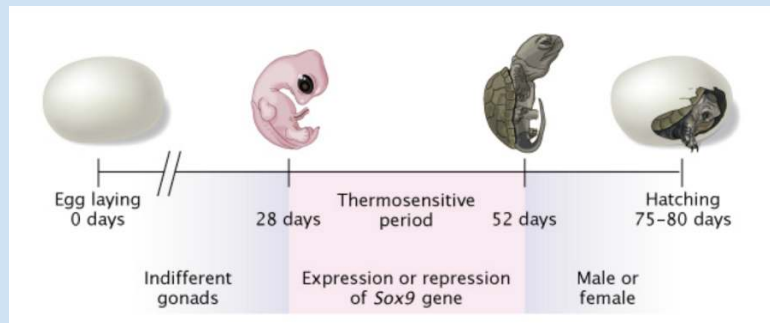
Hydrangeas

- Hydrangeas are common garden plants, and they may have pink or blue flowers depending on the pH of the soil
- If the soil pH is less than 5 (acidic), then the flowers will be blue
- If the soil pH is greater than 7 (alkali) then the flowers will be pink
- The acidity of the soil changes the availability of the aluminium ions in the soil, and this causes the colour change

Crocodiles and turtles

- Scientists know that the temperature at which a crocodile egg is incubated influences the sex of the resulting baby crocodile
- Studies have shown that there is a specific point in the development of certain organisms, known as the **thermosensitive period (TSP)**, during which the gonadal tissue is responsive to temperature. When exposed to one range of temperatures, this tissue develops into ovaries; when exposed to a different range, it develops into testes
- In the turtle *Emys orbicularis*: at incubation temperatures of 25°C, all such

- turtles are born male, but at temperatures of 30°C, all are born female
- Prior to the tsp, all of the embryonic gonads appear the same, however, during the TSP something happens to tell the tissues to become either ovaries or testes
 - That “something” is a change in gene expression that results in differentiation of the tissue to its new identity. In particular, expression of the Sox9 gene has been shown to change in response to temperature, with high expression levels at lower temperatures and repressed levels at higher temperatures. Because the transcription factor produced by Sox9 affects a gene that plays a major role in sex determination, the expression of this gene changes when temperature (and therefore Sox9 expression) changes, thus resulting in a specific phenotype of maleness or femaleness.



5.3.3- investigate the structure and function of proteins in living things

- Nearly every function of a living organism depends on proteins
- Proteins speed up chemical reactions, and play a role in cell-to-cell recognition and cellular communication, movement, storage, cell membrane functions and even structural support

Functions of Proteins

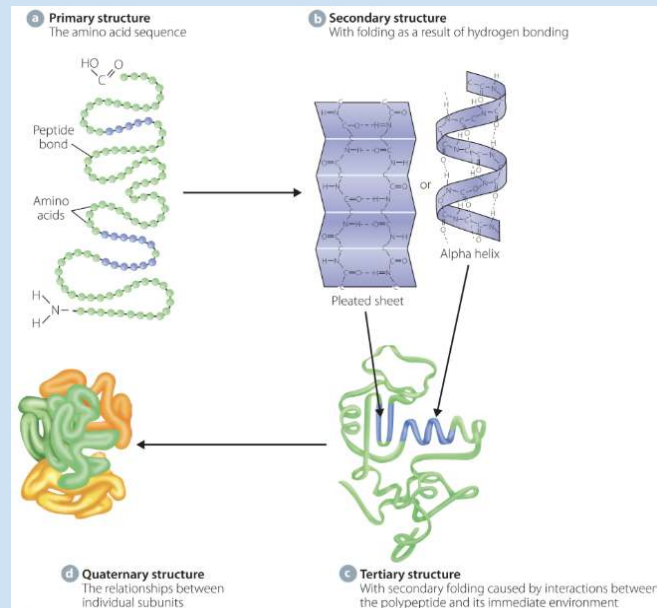
- The role of proteins as catalysts (enzymes) means they control reactions (metabolism) in cells
- Proteins control cell composition through their role in the cell membrane
- Transport proteins in body fluids such as haemoglobin in blood helps transport substances around the body
- Proteins strengthen cellular structures and enable movement by providing support, by interacting and by changing shape
- Many hormones are proteins that control processes such as reproduction, growth and the storage and release of glucose
- Proteins provide for storage, such as oxygen stored in ferritin
- Immune response results from the ability of some proteins to recognise other molecules

Protein Structure

- Proteins are large biomolecules and may be synthesised as one or from several polypeptide chains
- The chains are folded and organised into specific shapes that are vital to the correct functioning of the protein
- Most proteins are required to bind to other molecules to achieve functionality
- A single change to one amino acid in the sequence of polypeptides can alter the shape and therefore the function or functionality of the protein
- Proteins are made from 20 different types of amino acids

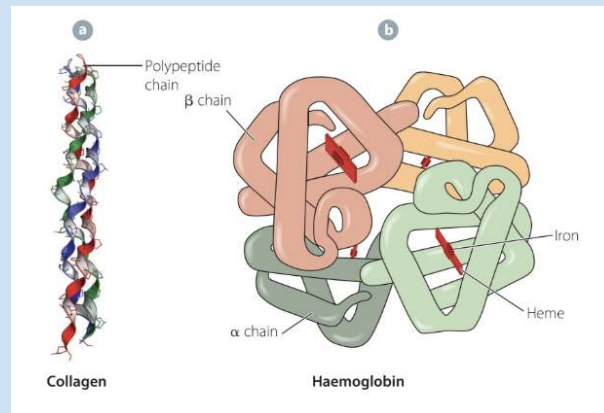
Levels of protein structure

- The structure of proteins can be described at four levels: primary, secondary, tertiary and quaternary
- Primary;
 - Basic structure of a protein is a polymer of amino acids, arranged in linear chains or polypeptides, and is termed its primary structure
- Secondary;
 - The secondary structure is the three-dimensional arrangement of the polypeptide chain. The secondary structure forms as a result of the amino acid chain becoming linked by hydrogen bonds, either twisting the polypeptide into a spiral (alpha helix) typical of a fibrous protein, or folding it into a pleated sheet rather than a spiral, also held together by fibrous bonds
- Tertiary;
 - Further holding leads to the tertiary structure that is seen in more complex proteins such as globular proteins. Certain forceps of attraction between alpha helices and pleated sheets (such as disulfide bonds) cause the polypeptide to fold into a more complex three-dimensional shape
- Quaternary;
 - Quaternary protein structure occurs in proteins that are made up of two or more polypeptide chains that link to create a more complex 3-dimensional structure



Types of proteins in cells

- Fibrous proteins form structural components of cells and tissues; together with water, they form the basic structure of protoplasm (the cytoskeleton). Fibrous proteins are long and insoluble in water.
- Collagen is an example of a long, stringy protein that is coiled and very strong. It is commonly found in skin, along with another fibrous protein, elastin. These proteins are also present in ligaments and tendons. Keratin is a fibrous protein in hair and nails.
- Globular proteins are usually spherical in shape and are compact and soluble in water. They are often transport proteins, such as haemoglobin in the blood. Immunoglobulins (antibodies), hormones and enzymes are other examples of globular proteins



Genetic Variation

5.4.1- conduct practical investigations to predict variations in the genotype of offspring by modelling meiosis, including the crossing over of homologous chromosomes, fertilisation and mutations

Gregor Johann Mendel (1822-1884)

- Mendel's Experiments
- Experimental Techniques
- Mendel's Conclusions
- Genes and Alleles
- Genotypes and Phenotypes
- Homozygous and Heterozygous
- Simple Dominance

Mendel's Experiments

- Studied the inheritance of different characteristics in pea plants (*Pisum sativum*)
- Mendel proposed a particulate hypothesis of inheritance. Discrete units (factors) are inherited from each parent. Pairs of these factors control inheritance
- Mendel identified 7 easily distinguishable characteristics (traits) that had alternative forms that he thought were controlled by factors (now known as genes)

Mendel's 7 traits

Trait	Alternative forms	
	Dominant	Recessive
Seed Shape	Round	Wrinkled
Flower Colour	Purple	White
Seed Colour	Yellow	Green
Pod Shape	Inflated	Constricted
Pod Colour	Green	Yellow
Stem Height	Tall	Short
Flower Position	Terminal (at end of stem)	Axillary (in leaf axil)

Experiments

- Mendel Bred his pea plants for two years to make sure that the plants to be used in his experiments were all true breeding (homozygous) so that when the plants were pollinated they produced plants that were identical to the parents (P)
- Under normal conditions pea plants are self-fertilising so Mendel artificially crossed the plants by removing stamens♂ from the flower buds and dusted the stigma ♀ with pollen that he had collected from another plant
- The first generation (F1) of plants were allowed to self-pollinate
- Mendel counted the numbers of offspring in the second generation (F2) with each characteristic
- He examined only individual characteristics

Experimental Techniques

- Mendel carried out his experiment from 1856-1863. His successes were a result of the following:
 - He was meticulous in recording his results and kept accurate records
 - Examined one trait at a time
 - Had a known history of the parentage of his plants and recorded it
 - Controlled the pollination process carefully
 - Used large numbers of plants and repeated his experiments many times
 - Kept an accurate record of the numbers of offspring and analysed his results mathematically
 - Chose peas which were easy to grow and produced offspring rapidly and could normally self-pollinate

Mendel's Conclusions

- Characteristics were not blended but were discrete units
- Each characteristic was controlled by a pair of factors
- The factors separated from one another when the sex cells were formed (Law of Segregation)

- At fertilisation, the offspring received one factor from each parent randomly (Law of Independent Assortment)
- One factor was dominant while the other was recessive
- Mendel found the Monohybrid Ratio of 3:1

Genes and Alleles

- Genes- section of a chromosome which controls the hereditary characteristics of an organism
- Genes consist of a DNA sequence that codes for a specific polypeptide
- Alleles- a pair of genes are called alleles which are alternative forms of the same gene
- The genes of a pair that occur on the same place on a homologous chromosome are called alleles
- Homologous chromosomes are matching pairs of chromosomes in a diploid organism

Homozygous and Heterozygous

- Homozygous- when the genes in the alleles are identical, they are called homozygous genotypes (pure breeding)
- Heterozygous - when the genes in the alleles are different they are called heterozygous genotypes (hybrid)

Simple Dominance

- Dominant genes are expressed (observable) whenever they are present
- When writing genotypes the dominant gene is written first with an uppercase letter
- Recessive genes are only expressed if both genes are the recessive form (no dominant gene present)
- When writing genotypes, the recessive gene is written last with a lowercase letter
- Notation:

Dominant gene → **Rr** ← Recessive gene

RR = Homozygous Dominant (pure/true breeding)

Rr = Heterozygous (Hybrid)

rr = Homozygous Recessive (pure/true breeding)

Example using Punnett Square

- Red Flower (R) is dominant over white flower (r)
 - A homozygous red (RR) is crossed with a homozygous white (rr)
- RR x rr**

P	R	R
r	Rr	Rr
r	Rr	Rr

F1 generation Genotype all Rr, Phenotype all red

- When members of the F1 generation are crossed the F2 generation is produced

Rr x Rr

F₁	R	r
R	RR	Rr
r	Rr	rr

F2 generation Genotypes RR, Rr, rr; phenotype 3 red: 1 white which confirms the red:white 3:1 Mendel ratio

Incomplete / Partial Dominance

- Occurs when neither allele shows dominance over the other
- This results in offspring with a phenotype different from either parent

Example: Breeding homozygous red snapdragon flowers (RR) with homozygous white snapdragon flowers (WW)

RR x WW

P	R	R
W	RW	RW
W	RW	RW

F1 Generation

- The genotype of the offspring are all RW,
- The phenotype of the offspring are all pink
- When the F1 generations are crossed, the following results are obtained:

F ₁	R	W
R	RR	RW
W	RW	WW

F₂ Generation

Genotype	RR	:	RW	:	WW
Phenotype	Red	:	Pink	:	White
Ratio	1	:	2	:	1
Percentage	25%	:	50%	:	25%

Incomplete dominance occurred in 50% of the F₂ as pink snapdragons

Co-Dominance

- Is another way how ratios vary from Mendel's ratios
- When Co-dominance occurs, both alleles are expressed in the phenotypes of the offspring

Example: A white haired cow (WW) is crossed with a red haired bull (RR)
RR x WW

P	W	W
R	RW	RW
R	RW	RW

F₁ generation genotype - all RW,

phenotype all strawberry roan

- The phenotype "strawberry roan" is a combination of red hairs and white hairs on the coat of the animal

Sex linkage

- Occurs when the genes are carried on either sex determining chromosome
- The sex determining chromosomes exist as the 23rd pair of human chromosomes in the form of X and Y chromosomes
- In female cells, the 23rd pair consists of two X chromosomes (XX)
- In male cells, the 23rd pair consists of one X and one Y chromosome (XY)

Determining sex of children

- Using a punnett square

P	X	X
X	XX	XX
Y	XY	XY

F1 Generation

Genotype	XX	:	XY
Phenotype	female	:	male
Ratio	1	:	1
Percentage	50%	:	50%

- There is an equal chance of producing male or female offspring

Sex-linked disease

- When a gene is carried on an X or Y chromosome, various results can be obtained.
- Example: the gene for colour blindness is carried on the X chromosome (X^c)
- The following phenotypes may occur on a cross containing an X^c

Genotype	Phenotype
XX	Normal female
X^cX	Normal female (carrier)
X^cX^c	Colour blind female
XY	Normal male
X^cY	Colour blind male

Pedigrees

- The inheritance of characteristics in a family line can be shown by using a pedigree
- A pedigree can display either sex-linked characteristics

Biology Excel Extra Notes;

- Mutations are the ultimate source of genetic variation
- Punnett square diagrams show the possible genotype of gametes and possible combinations on fertilisation
- Each fertilisation event is independent of any other

Male gametes→ Female gametes↓	T	t
T	TT	Tt
t	Tt	tt

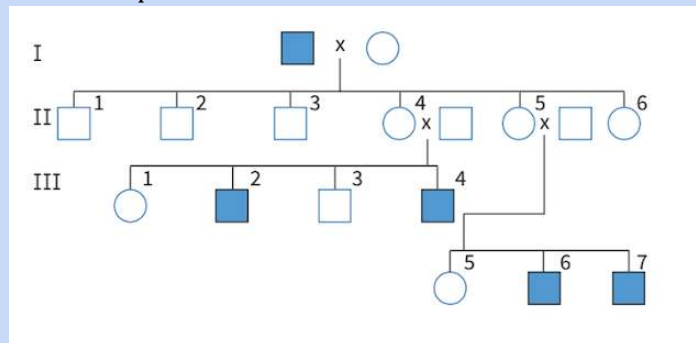
5.4.2- Model the formation of new combinations of genotypes produced during meiosis, including but not limited to -

- Interpreting examples of autosomal, sex linkage, co-dominance, incomplete dominance, and multiple alleles
- Constructing and interpreting information and data from pedigrees and Punnet squares

Biology Excel Notes;

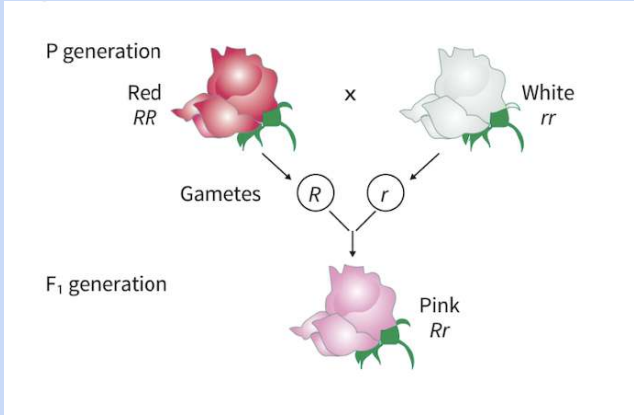
Autosomal inheritance and sex linkage

- **Autosomal inheritance** is the inheritance of genes carried on normal (autosomal) chromosomes that impacts males and females equally
- **Alleles can interact in a dominance/recessive relationship** in which the dominant allele is expressed in the hybrid (heterozygous) form and the recessive phenotype is only expressed in the homozygous genotype
- **Sex chromosomes**, such as the X and Y chromosomes in humans, determine the sex of offspring, but the X chromosome also carries some genes which are expressed more frequently in males that only have one X
- The pattern of sex-linked inheritance is different in males to female



Co-dominance, incomplete dominance, and multiple alleles

- Alleles can interact in a **co-dominant** manner in which both alleles are expressed (e.g, roan cattle)
- In **incomplete dominance**, the heterozygous form has an intermediate or a blended phenotype, different from either parent
- **Multiple allele** is the inheritance of a characteristic in which there are more than two forms of alleles so there is a great variety of possible phenotypes and genotypes (i.e, human blood types may be A or B - co-dominant - or O - recessive to both A and B)

	 The diagram illustrates a monohybrid cross. In the P generation, a red flower with genotype RR is crossed with a white flower with genotype rr. Arrows labeled 'Gametes' point to the alleles R and r. These combine in the F₁ generation to form a pink flower with genotype Rr.
5.4.3- Collect, record and present data to represent frequencies of characteristics in a population, in order to identify trends, patterns, relationships, and limitations in data, for example; - Examining frequency data - Analysing single nucleotide polymorphism (SNPs)	Biology Excel Notes; Frequency data of characteristics in populations • Genetic diversity is reflected in the frequency of characteristics across a population • Frequences of characteristics are collected, recorded and presented to reveal data on phenotypes in local populations - Observations could include characteristics such as eye colour, hair colour, tongue-rolling, or attached or detached ear lobes • Data is limited because phenotypes do not always indicate genotype • Sampling may be insufficient to reflect the whole population gene frequency • Allele frequency is a measure of the genetic diversity of a gene pool Analysing single nucleotides polymorphisms • A single nucleotide polymorphism (SNP) is an alternative form of an allele resulting from a single base substitution (mutation) in the DNA which occurs in at least 1% of the population • HbS is a SNP of normal haemoglobin (HbA) • The sickle-cell trait is a result of HbS in the heterozygous form • Sickle-cell anaemia is the result in the homozygous form • 90% of human variation in DNA occurs as SNPs are found in non-coding and coding sections of DNA
Inheritance Patterns in a Population	
5.5.1- Investigate the use of technologies to determine inheritance patterns in a population, using, for example: - DNA sequencing and profiling	DNA sequencing • The process of determining the sequence of nucleotide bases in a piece of DNA • Tells scientists the kind of genetic information that is carried in a particular DNA segment - Sequence information can be used to determine which stretches of DNA contain genes and which stretches carry regulatory instructions, turning

genes on or off

- Requires breaking the DNA of the genome into many smaller pieces, sequencing the pieces, and assembling the sequences into a single long “consensus”

Technologies;

- Polymerase chain reaction (PCR) (used in combination with sequencing)
- Sequencing by synthesis (SBS)
- Sequencing by oligonucleotide ligation and detection (SOLiD)
- Labelling (i.e, use of fluorescent dyes)
- Separating by electrophoresis
- Base detection (i.e, use of dyes or light emission)
- Examples of use include detection of SNPs as indicators of genetic diversity and disease risk and in the Human Genome Project

The Maxam-Gilbert Method

1. Isolate DNA strand
2. Radioactively label one end of DNA by adding a radioactive phosphorus atom to the phosphate molecule
3. Chemicals and conditions are then modified to suit a specific base which results in the base being removed through the ribose sugar it is attached to
4. The DNA strand is cleaved (cut) at this site. This occurs at each position of that specific base in the DNA strand being sequenced
5. The DNA fragments formed then undergo gel electrophoresis, with the shorter strands moving further through the gel than the longer strands
6. When all patterns from the gel electrophoresis are compared, the sequence of the bases on the DNA strand can then be determined

DNA Profiling

- DNA profiling = the use of DNA tests to establish identity or relationships
- Most individuals of the same species have nearly identical DNA, but the sequence at certain locations (loci) throughout the genome varies among individuals
 - Variations can be used to distinguish one individual from another
- In profiling;
 - DNA is taken from a biological sample
 - PCR (polymerase chain reaction) is used to produce copies of STR sequences
- Examples of use;
 - Determining the probability of identity (such as matching crime scene evidence to individuals)
 - Or a familial relationship by comparing highly variable parts of the DNA of individuals

The Process (Electrophoresis separation example)

1. Obtain a sample of DNA (saliva, epithelial cells)
2. Extract the DNA - chemicals are added to break open and extract the DNA
3. Copy DNA - STRs (short tandem repeats) are sections of non-coding DNA that are repeated

	4. Determine the size of the STRs - The size of STRs at each genetic locus is determined using a genetic analyser. The genetic analyser separates the copied DNA by gel electrophoresis and can detect the fluorescent dye on each STR. This is the same piece of equipment used for sequencing 5. Match - the number of times a nucleotide sequence is repeated in each STR can be calculated from the size of the STRs
5.5.2- Investigate the use of data analysis from a large-scale collaborative project to identify trends, patterns, and relationships, for example; - The use of population genetics data in conservation management - Population genetics studies used to determine the inheritance of a disease or disorder - Population genetics relating to human evolution	Conservation Management ● Population genetics investigates factors that influence the change of frequency of alleles over time and across locations and populations ● Allele frequency changes with mutations, natural selection, gene drift and gene flow ● Single nucleotide polymorphisms (SNPs) are used as an indicator of levels of genetic diversity and to compare the genetic diversity of species ● Sample size is important for obtaining reliable data for studying population genetics ● Genetic diversity is important for the survival of endangered species and small or isolated populations ● Conservation management needs to consider genetic diversity as well as habitat requirements ● Tasmanian devils are at risk from a contagious cancer that is thought to be linked to the low diversity of genes associated with its immune system - Devil Facial Tumour Disease (DFTD) -> makes it near impossible for them to eat and has resulted in the species becoming endangered Inheritance of a disease or disorder and human evolution ● Many diseases have both genetic and environmental causes so population genetics data can help determine the genetic factors that contribute ● Genome-wide association (GWA) studies have helped to collect and analyse data on the genetic components of disease ● The frequency of alleles for sickle-cell anaemia resulting from an SNP that changes haemoglobin is higher in areas where there is exposure to malaria as it provides some resistance to the protozoan pathogen ● In human evolution, mtDNA is used as a molecular clock to determine the data of the last common ancestor ● DNA from mitochondria and the nucleus are used to compare modern humans with recent ancestors such as Neanderthals and Denisovans